

Mechanical Power Module (MPM) Measurement Framework

Takeaway: This framework defines *what*, *how*, and *why* the MPM is measured across engineering, logistics, safety, economics, and interoperability domains. It is structured like a standards-grade measurement system suitable for pilot deployments, consortium compliance, and long-term industrial scaling.

1. Measurement Philosophy

The MPM is a **universal mechanical-power primitive**, so its measurement framework must:

- Quantify **interface performance**, not just module performance
- Capture **logistics repeatability** across operators and environments
- Validate **safety envelope integrity** under real-world duty cycles
- Support **multi-vendor interoperability**
- Produce **standards-ready data** for MPM v1.0 and future revisions

Every metric is tied to a specific compliance requirement in the emerging standard.

2. Measurement Domains

Each domain begins with a Guided Link so you can explore or expand it.

A. Mechanical Performance

Measures how effectively the MPM delivers rotational mechanical power.

- **Torque Delivery Stability** — RMS deviation across duty cycles
- **Speed Regulation** — % variance under load transitions
- **Thermal Behavior** — temperature rise per kW delivered
- **Vibration Envelope** — mm/s RMS across mounting points
- **Mechanical Efficiency** — input vs. delivered mechanical output

Primary Instruments: torque transducers, tachometers, thermal sensors, vibration probes.

B. Interface Integrity

Measures the health and reliability of the standardized mechanical interface.

- **Spline Wear Rate** — μm wear per operating hour

- **Axial Retention Force** — kN required for controlled disconnect
- **Alignment Tolerance Drift** — mm deviation over time
- **Latch-Cycle Durability** — cycles to first measurable degradation

Primary Instruments: optical metrology, force gauges, alignment jigs.

C. Safety Envelope Performance

Ensures the MPM meets its safety requirements under all operating conditions.

- **Interlock Reliability** — % successful engagement/disengagement
- **Emergency Stop Response Time** — ms from trigger to mechanical isolation
- **Uncontrolled Disconnect Incidents** — target: zero
- **Thermal Overrun Events** — count + severity index

Primary Instruments: interlock telemetry, high-speed logging, thermal sensors.

D. Logistics & Swap-Out Performance

Measures how well the MPM functions as a *logistics primitive*.

- **Swap-Out Time** — target: <15 minutes
- **Operator Variability** — time variance across trained operators
- **Transport Shock Exposure** — g-force profile during movement
- **Storage Compliance** — adherence to environmental envelope

Primary Instruments: time logs, shock sensors, environmental monitors.

E. Host-System Interoperability

Measures cross-platform compatibility.

- **Successful Host Connections** — % of attempts across host types
- **Performance Degradation Across Hosts** — torque/speed delta
- **Adapter Wear Rate** — μm wear per cycle
- **Host-Side Safety Events** — count + severity

Primary Instruments: host telemetry, interface metrology.

F. Economic Performance

Quantifies the economic value of modular mechanical power.

- **Lifecycle Cost Reduction** — % vs. fixed mechanical systems

- **Maintenance Time Reduction** — hours saved per module
- **Inventory Efficiency** — modules per operational node
- **Vendor Interoperability Savings** — cost delta from multi-vendor sourcing

Primary Instruments: cost models, maintenance logs, procurement data.

3. Measurement Architecture

A standards-grade measurement system requires consistent architecture.

A. Telemetry Layer

- Real-time mechanical data
- Safety-interlock state
- Interface-health indicators
- Compliance logging

B. Data Schema

- Time-series mechanical data
- Event-based safety logs
- Geometry & wear measurements
- Logistics timestamps

C. Data Quality Requirements

- Timestamp accuracy: ± 1 ms
- Sensor calibration: NIST-traceable
- Wear measurement resolution: ≤ 5 μm
- Environmental logging: 1-minute intervals

4. Measurement Cadence

Continuous

- Torque, speed, thermal, vibration
- Interlock state
- Host-system telemetry

Per Swap-Out

- Alignment check
- Latch-cycle count

- Operator time log

Weekly

- Interface wear measurement
- Safety envelope review

Monthly

- Logistics performance summary
- Host-system interoperability report

Quarterly

- Standards refinement proposals
- Consortium compliance updates

5. Compliance Thresholds

These thresholds define “MPM-compliant” performance.

- Torque stability: $\pm 3\%$
- Speed regulation: $\pm 2\%$
- Thermal rise: $< 15^{\circ}\text{C}$ over baseline
- Vibration envelope: $< 4 \text{ mm/s RMS}$
- Interlock reliability: $> 99.98\%$
- Swap-out time: < 15 minutes (95th percentile)
- Wear rate: $< 10 \text{ }\mu\text{m}$ per 100 hours
- Zero uncontrolled disconnects

6. Reporting Structure

A. Engineering Report

- Mechanical performance summary
- Interface integrity analysis
- Safety envelope validation

B. Logistics Report

- Swap-out metrics
- Transport shock profile
- Operator variability

C. Interoperability Report

- Host-system compatibility
- Adapter wear
- Cross-platform performance deltas

D. Standards Update Packet

- Proposed geometry refinements
- Updated tolerances
- Revised safety requirements

7. Success Criteria

The MPM measurement framework is successful when:

- Data is **repeatable**, **cross-vendor**, and **standards-grade**
- Measurements directly inform **MPM Standard v1.0**
- Telemetry supports **real-time compliance**
- Logistics metrics validate the MPM as a **mechanical logistics primitive**
- Interoperability metrics confirm **universal mechanical**